Partial Differential Equations (PDE)
Assignment 5

[Series solution]

(a) Show that the solution of the boundary value problem

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & \text{for } 0 < x < L \\ u(0,t) = 0 = u(L,t) \end{cases}$$

with the initial conditions

$$u(x,0) = \phi(x) \qquad u_t(x,0) = \psi(x)$$

is

$$u(x,t) = \sum_n \left(A_n \cos \frac{n\pi ct}{L} + B_n \sin \frac{n\pi ct}{L} \right) \sin \frac{n\pi x}{L}$$

where A_n and B_n satisfy that

$$A_n = \frac{2}{L} \int_0^L \phi(x) \sin \frac{n\pi x}{L} dx \qquad \text{and} \qquad B_n = \frac{2}{n\pi c} \int_0^L \psi(x) \sin \frac{n\pi x}{L} dx.$$

[Hint]

$$\int_0^L \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = \frac{L}{2} \delta_{n,m}$$

(b) Show that the solution of the boundary value problem

$$\begin{cases} u_t - a^2 u_{xx} = 0, & \text{for } 0 < x < L,\ 0 < t < \infty \\ u(0,t) = u(L,t) = 0 \\ u(x,0) = \phi(x) \end{cases}$$

is

$$u(x,t) = \sum_{n=1}^{\infty} A_n e^{-\frac{n^2\pi^2}{L^2} a^2 t} \sin \frac{n\pi x}{L}$$

where A_n satisfies that

$$A_n = \frac{2}{L} \int_0^L \phi(x) \sin \frac{n\pi x}{L} dx.$$